

Sai Tejas Repala

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Final-year IT undergraduate specializing in Machine Learning, Deep Learning, and Generative AI — with hands-on experience building RAG pipelines, LLM-powered applications, and predictive analytics models. Skilled in end-to-end AI development from data preprocessing and model training to full-stack deployment using React.js, Node.js, and AWS. Passionate about building intelligent systems that solve real-world problems at scale.

EDUCATION

Manipal Institute of Technology

B.Tech in Information Technology; CGPA: 8.36

Bangalore, India

Aug. 2023 – May 2027

EXPERIENCE

IoT Intern

May 2026 — June 2026

Vallynx Solutions Pvt Ltd

Remote

- Engineered a **secure IoT communication layer** using **CoAP**, **DTLS-PSK**, Eclipse Californium 3.x, and Scandium on **AWS EC2**, enabling encrypted machine-to-machine (**M2M**) communication with minimal protocol overhead for resource-constrained, low-power devices.
- Deployed and configured a **production-grade CoAP** server on AWS EC2 (Amazon Linux) with Elastic IP, Security Group rules, and background process management — providing globally accessible, **cloud-hosted IoT** endpoints.
- Documented the CoAP/DTLS communication workflow and AWS EC2 **deployment architecture**, creating **technical handoff artifacts** including user stories, acceptance criteria, and implementation notes.

Frontend Intern

June 2025 – July 2025

Ada Insis

Hyderabad, India

- Engineered 2 production-deployed React.js applications (**Voter Management System & Electricity Bill Dashboard**) with role-based admin dashboards, and end-to-end from architecture to deployment.
- Maintained **code quality** across a collaborative codebase using **Git workflows** (branching, PRs, conflict resolution) in a **simulated production environment**.

PROJECTS

Multi-PDF RAG Chatbot | *Python, LangChain, Pinecone, Gemini API, Hugging Face, Sentence Transformers*

- Developed a **Retrieval-Augmented Generation (RAG)** chatbot using **Python**, **LangChain**, **PineCone**, and the **Gemini API** to answer questions from multiple PDF documents with context-aware responses.
- Implemented **PDF parsing**, **text chunking**, and semantic embeddings using **Hugging Face Sentence Transformers**, enabling efficient semantic retrieval through the **PineCone vector database**.
- Built an interactive **Streamlit** UI featuring **conversation history** and **source citations** for accurate outputs.

AI-Powered Facial Recognition Attendance System | *React, TensorFlow.js, face-api.js, Supabase*

- Developed a full-stack smart attendance and anti-spoofing portal utilizing **React**, Vite, Tailwind CSS, and **face-api.js** for **real-time edge facial recognition**.
- Engineered an anti-spoofing **liveness verification** engine incorporating landmark aspect ratio computations (**EAR/MAR**) and **confidence boundary filters** to detect and block static photo/screen presentation attacks.
- Architected a **dual-storage** data persistence layer via **Supabase** and a robust localStorage **offline fallback**.

TECHNICAL SKILLS

Programming : Java, C, Python, JavaScript, SQL

Machine Learning : Scikit-learn, Random Forest, Decision Tree, Logistic Regression, Linear Regression, Support Vector Machine, Clustering Techniques

Deep Learning : TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Transformers

Generative AI & NLP : LangChain, RAG, Pinecone, Gemini API, Hugging Face, Sentence Transformers, TF-IDF, Cosine Similarity, Named Entity Recognition

Frameworks : React.js, Node.js, NumPy, Pandas, Scikit-Learn, Streamlit

Databases & Tools : Git, GitHub, VS Code, Jupyter Notebook, Oracle, MySQL, MongoDB

Core Concepts : Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS & CO-CURRICULAR

Certifications: Machine Learning with Python, Neural Network and Deep Learning, Generative AI Foundations.

Achievements: 24-hour ACM Hackathon Participant; Gold Medal in Intercollegiate Volleyball tournament.